

Automated Support for Flexible Extensions

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Google Tool with Email/LMS Management

Manages “soft” deadlines by automatically emailing students regarding short extensions, and updating Gradescope with student-specific deadlines.

- 20,000 students over 10+ courses over 2 years
- In Spring 2024, at least 10 courses are using the tool including 2 lower division courses and 8 upper division courses across the CS and DS departments

	CS 161 Capacity	Number of Form Submissions for CS 161	Number of Unique Students Who Submitted to the Form	Percentage of Course Who Requested an Extension
Fall 2022	618	2300	449	72.65%
Spring 2023	751	4055	700	93.21%
Summer 2023	290	1243	210	72.41%
Fall 2023	690	4153	618	89.57%
Spring 2024	625	1940 so far	529 so far	84.64%

How many days would you like an extension for? *

Please enter the total number of days between the original due date and your requested due date. If you already have an extension on the assignment, please still enter the total number of days. Our system uses absolute offsets, and adding is hard! If you selected multiple assignments above, please comma-separate the number of days for each assignment request.

Your answer

Why do you need this extension? *

Your answer

Are you working with one (or more) partners on this assignment? *

Your extension will apply to them as well.

- ☐ Yes
- ☐ No



Google Form students submit to request an extension.

	B	C	D	E	M	N
	is_dsp	proj1	multi1	proj2	approval_status	email_status
@berkeley.edu	1	4	1	4	Manually Approved	Manually Sent
ley.edu	1				Auto Approved	Auto Sent
berkeley.edu	1					
ey.edu	1					
ey.edu	1	2	2		Auto Approved	Auto Sent
ey.edu	1	2	2	3	Auto Approved	Auto Sent
berkeley.edu	1					
@berkeley.e	1	2	2	3	Auto Approved	Auto Sent
berkeley.edu	1	2		4	Manually Approved	Auto Sent
y.edu	1	3	3	4	Manually Approved	Auto S
@berkeley.edi	1	3	3	4	Manually Approved	Auto S
ey.edu	1		1		Auto Approved	Auto S
@berkeley.ec	1			3	Auto Approved	Auto S
berkeley.ed		3			Manually Approved	Manu
ey.edu		4			Manually Approved	Auto S



Google Sheets roster of students

Extensions					
Assignment Settings					
Release Date (PST)	Due Date (PST)	Late Due Date (PST)			
Sep 08 2023 11:59 PM	Sep 21 2023 05:00 PM	Sep 24 2023 05:00 PM			
Student Extensions					
First & Last Name	Extension Type	Sections	Release (PST)	Due (PST)	Late Due (PST)
	Due Date				
	Late Due Date				
	Due Date				
	Late Due Date				
	Due Date				
	Late Due Date				

Gradescope extensions created automatically by the tool

Student and Google Form

- Asks name, SID, (optionally) Disabled Students Program status, assignments, length of extension request, why they are requesting an extension, if they have a partner and who that partner is

Tool and Instructor/Staff

- Three main functions hosted by Google Cloud: handle_form_submit, handle_email_queue, handle_flush_gradescope
- Extension requests auto populate from the Google Form into one row per student email
- Based on given variables, will automatically send an email to students that their request has been approved
- Manually review non automatically approved requests, approve them, and send automated emails

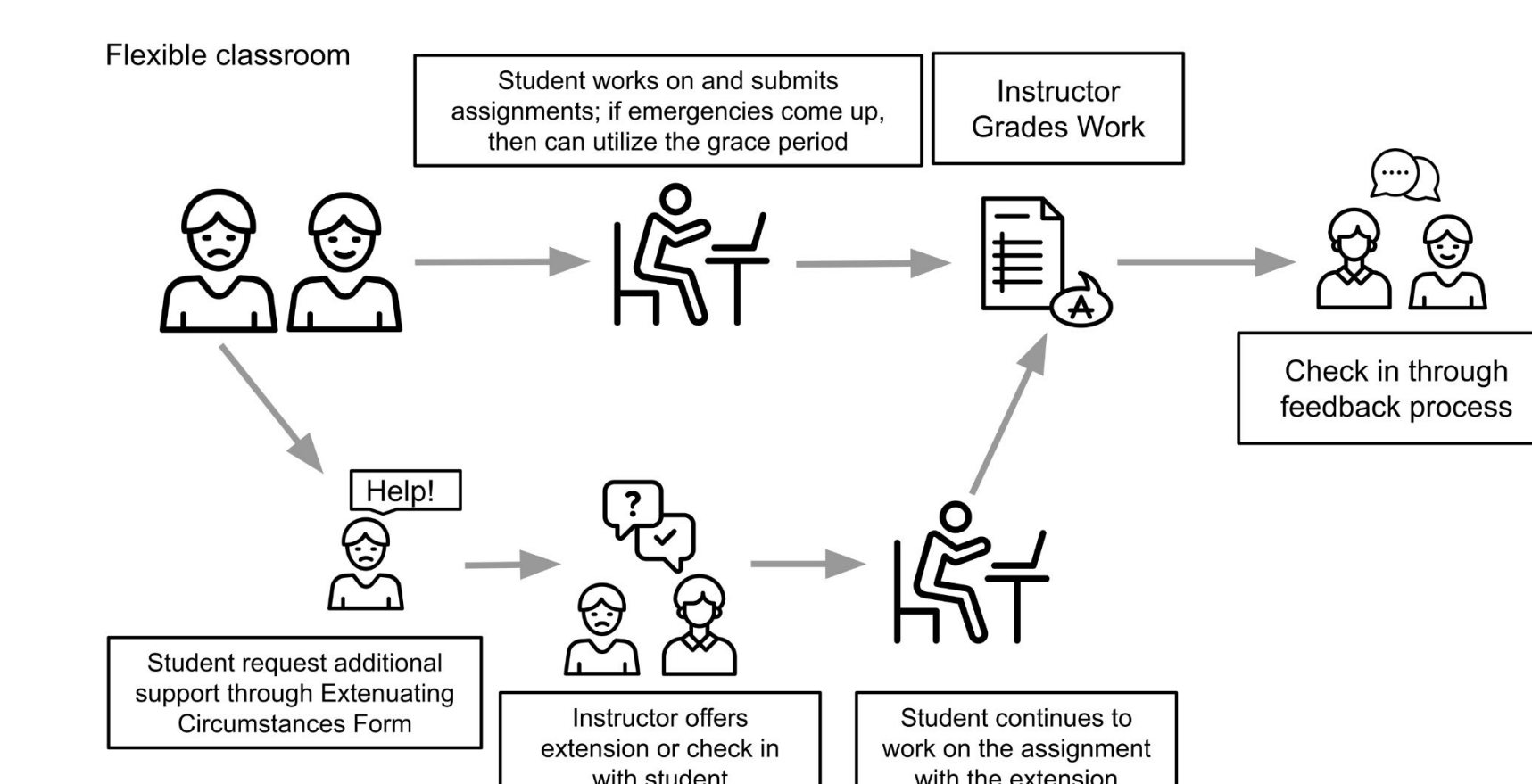
	A	B	
1	key	value	description
2	EMAIL_FROM	CS 161 Staff <example@email.edu>	One email, formatted as "Some Name <email@berkeley.edu>"
3	EMAIL_CC	example@email.edu	Zero or more comma-separated emails to be CC'd.
4	EMAIL_SIGNATURE	EvanBot	The email's signature
5	EMAIL_SUBJECT	[CS 161] Extension Request Update	The email's subject line
6	EMAIL_REPLY_TO	example@email.edu	The email's reply-to address (usually, the DSP/Extensions TA)
7	SLACK_ENDPOINT	TOD0	Your Slack webhook for the extensions-bot channel
8	SLACK_DEBUG_ENDPOINT	TOD0	Do not touch - used for debugging purposes (error logs are)
9	AUTO_APPROVE_THRESHOLD	3	The maximum number of days to auto-approve extension requ
10	AUTO_APPROVE_THRESHOLD_DSP	7	The maximum number of days to auto-approve extension requ
11	AUTO_APPROVE_ASSIGNMENT_THRESHOLD	3	The maximum number of (form-wise per submission) extension
12	MAX_TOTAL_REQUESTED_EXTENSIONS_THRE	1	The maximum number of (global per student) extensions to aut
13	SLACK_TAG_LIST	U02BR9TPNPK	Zero or more comma-separated list of Slack User ID's you wan
14	SPREADSHEET_URL	https://docs.google.com/spreadsheets/d/1Bab1	Paste the URL of this spreadsheet here!!!
15	EXTEND_GRADESCOPE_ASSIGNMENTS	No	Do you want to enable the Gradescope assignment extension f
16	GRADESCOPE_EMAIL	TOD0	Email of a gradescope staff account with permission to manage
17	GRADESCOPE_PASSWORD	TOD0	Password of a gradescope staff account with permission to ma

Google Sheets environment variables

Next Steps

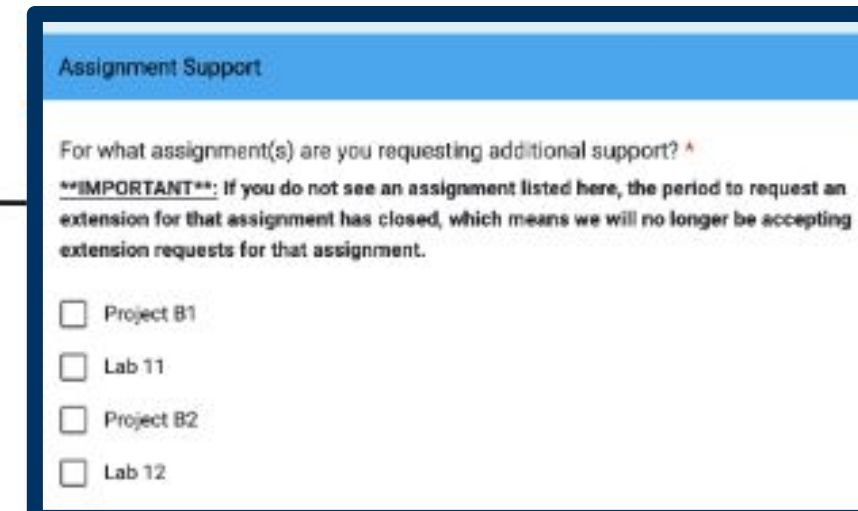
- **LMS support:** Currently, Gradescope extensions can be automatically instantiated. We hope to incorporate Canvas next.
- **Automated Email:** Currently, the automated email template is not editable. We hope to change this so the template is flexible per course.
- **Error handling:** We hope to incorporate better error handling.

Relevance

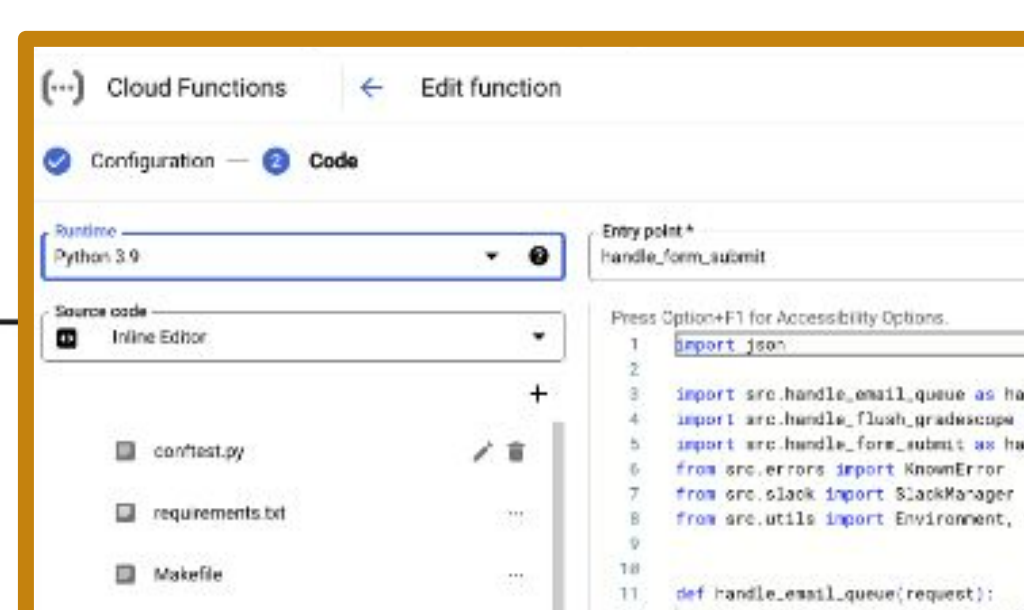


- Mastery learning approach
- Allow students to master concepts more thoroughly, even if it means that they would need more time

Student:
Student requests support through Google form



Tool:
Sorts student response into a spreadsheet, sends emails for auto-approved requests.



Instructor/staff:
Sets extension parameters, sends emails for manually approved requests.

Timestamp	What's going on	Which assignme
0 1/22/2023 15:23	Lorem ipsum do	Lab 1, Homework
1 1/22/2023 15:23	Lorem ipsum do	Lab 1, Homework
2 1/23/2023 15:57	Lorem ipsum do	Discussion Atter
3 1/23/2023 16:20	Lorem ipsum do	Lecture Participa
4 1/24/2023 11:52	Lorem ipsum do	Lab 1, Homework
5 1/24/2023 21:18	Lorem ipsum do	Discussion Atter

